

Xudong Wu

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🔗 [Agentic-Intelligence-Lab/EnergyBridge](#)

PhD student working on reinforcement learning, LLM alignment, agentic AI systems, and post-training.

Education

The University of Hong Kong (HKU)

Doctor of Philosophy

Hong Kong SAR

10/2025 – 09/2029 (Expected)

Supervisors: Prof. Jiayu Chen.

University of Edinburgh

BSc (Hons) Mathematics and Statistics

Edinburgh, UK

09/2023 – 07/2025

Average Score: 78/100, First Class Honours.

Dissertation: Amortized Inference.

Dalian University of Technology

BSc Information and Computing Science

Dalian, China

09/2021 – 06/2023

Overall GPA: 3.99/4.00, Top 5%.

Selected Publications and Preprints

- **Xudong Wu**, Pangpang Liu, Vaneet Aggarwal, and Jiayu Chen. “On the Convergence of Self-Improving Online LLM Alignment.” *UAI*, 2026. Accepted.
- Aravind Venugopal, Jiayu Chen, **Xudong Wu**, Chongyi Zheng, Benjamin Eysenbach, and Jeff Schneider. “Occupancy Reward Shaping: Improving Credit Assignment for Offline Goal-Conditioned Reinforcement Learning.” *ICLR*, 2026. Accepted.
- **Xudong Wu**, Jian Qian, Pangpang Liu, Vaneet Aggarwal, and Jiayu Chen. “Distributionally Robust Listwise Preference Optimization.” Under review at *NeurIPS*, 2026.
- Haixiang Sun, Yang Xu, Jiefu Zhang, **Xudong Wu**, Zihan Zhou, Jun He, and Jiayu Chen. “Core-Halo Decomposition: Decentralizing Large-Scale Fixed-Point Problem.” Under review at *NeurIPS*, 2026.
- **Xudong Wu**, Pangpang Liu, Vaneet Aggarwal, and Jiayu Chen. “SAIL-TRPO: A Trust-Region Online RLHF Algorithm with Guaranteed Fast Convergence.” Manuscript in preparation.
- Jiayu Chen, **Xudong Wu**, Zhekai Wang, and Vaneet Aggarwal. “Hierarchical Deep Counterfactual Regret Minimization.” Manuscript in preparation.

Research Experience

EnergyBridge: Long-Horizon LLM Agents for Virtual Power Plants

HKU

Project Lead, Agentic Intelligence Lab; project in progress

05/2026 – Present

- Lead the development of an end-to-end LLM-agent system for home-grid coordination, combining long-horizon planning, tool-augmented reasoning, persistent memory, preference learning, safety guardrails, and closed-loop execution.
- Design the full agent loop from observation to action: home/grid state perception, context retrieval, memory grounding, multi-agent negotiation, plan generation, tool/API invocation, constraint checking, device actuation, feedback collection, and memory update.
- Build preference-aware household agents and grid-aware coordinator agents for personalized demand response, enabling agents to optimize comfort, electricity cost, appliance schedules, VPP compliance, interruption cost, and user-specific constraints.
- Develop agent evaluation and self-improvement infrastructure with persona simulation, human-in-the-loop feedback, trajectory logging, reward/objective scoring, failure diagnosis, scenario replay, and repeated-interaction preference adaptation.
- Compare LLM-agent policies against rule-based agents, MPC, EnergyPlus-informed MPC, and RL-based controllers under realistic demand-response events, using metrics for task success, safety, comfort, user satisfaction, grid compliance, and adaptation quality.

Self-Play LLM Agent Training with GRPO and PSRO

HKU

Researcher, Agentic Intelligence Lab

2026 – Present

- Work on self-play post-training for LLM agents, combining Group Relative Policy Optimization (GRPO), policy rollouts, reward/verifier signals, and Policy-Space Response Oracles (PSRO).
- Formulate agent training as population-based reinforcement learning, where policies are iteratively optimized against evolving opponent, environment, and task distributions rather than static supervised data.
- Explore scalable RLHF/RLAIF-style pipelines for agentic reasoning, including trajectory sampling, outcome-based rewards, verifier-guided policy improvement, opponent sampling, and held-out strategic evaluation.
- Focus on improving strategic reasoning, robustness, tool-use behavior, and multi-agent adaptation through game-theoretic self-play and reinforcement learning post-training.

RLHF, Preference Optimization, and Online Alignment Theory

HKU

Researcher, Agentic Intelligence Lab

2025 – Present

- Study theoretical foundations of online LLM alignment, including self-improving preference optimization, policy improvement dynamics, stability, and convergence under model-generated feedback.
- Develop robust RLHF algorithms beyond pairwise preference learning, including distributionally robust listwise preference optimization under preference noise, distribution shift, and ranking uncertainty.
- Work on trust-region online RLHF methods with controlled policy updates, connecting practical post-training algorithms with classical reinforcement learning theory and convergence guarantees.

Reinforcement Learning, Offline RL, and Game-Theoretic Optimization

HKU / Collaborators

Collaborating Researcher

2025 – Present

- Contribute to projects on offline goal-conditioned RL, occupancy-based reward shaping, hierarchical counterfactual regret minimization, and large-scale fixed-point problem decomposition.
- Focus on algorithmic formulation, proof checking, theoretical positioning, and translating RL/game-theoretic ideas into publishable research claims.

LLM Agents for Automated Workflow Analysis on the Texera Platform

University of California, Irvine

Research Assistant, Donald Bren School of ICS

06/2024 – 09/2024

- Worked on Texera, a large-scale workflow engine for data analytics, with emphasis on LLM-assisted workflow interpretation, backend integration, HTML report generation, and real-time visualization.
- Designed a modular LLM-agent reporting component to interpret pipeline outputs, generate structured summaries, and support natural-language workflow analysis.
- Implemented system-level improvements for interactive report generation and scalable execution in a workflow-oriented data-analysis platform.

Selected Project Experience

Self-Rewarding Framework for Medical LLMs

University of Edinburgh

Project Lead, Machine Learning Practical

2025

- Designed a self-rewarding alignment framework for medical LLMs using DPO-style preference optimization and dynamically refined LLM-as-a-judge prompts.
- Built an evaluation pipeline with self-question generation, multi-response sampling, self-judgment scoring, and reward refinement on PubMedQA and MedMCQA-style tasks.
- Analyzed failure modes caused by reward misspecification, hallucination accumulation, and task difficulty, highlighting limits of closed-loop self-improvement.

A Comparative Study of Simulation-Based Inference Algorithms

University of Edinburgh

Honours Dissertation, Advisor: Dr. Amanda Lenzi

2024 – 2025

- Conducted a comparative study of amortized simulation-based inference methods, including BayesFlow, Sequential Neural Likelihood, and Affine Flow Matching.
- Designed controlled benchmark experiments for posterior recovery and calibration under structured statistical models, with emphasis on inference quality, robustness, and diagnostic evaluation.
- Extended the study to spatial disease-mapping settings with Poisson-CAR models, analyzing posterior calibration, parameter recovery, and robustness under strong spatial dependence.

Honors and Awards

- First Class Honours, University of Edinburgh, 2025.
- Outstanding Student of Dalian University of Technology, 2021–2023.
- First-Class Scholarship of Dalian University of Technology, 2022–2023.
- International Study Scholarship of Dalian University of Technology, 2022–2023.
- Second-Class Scholarship of Dalian University of Technology, 2021–2022.

Skills

Research: Reinforcement learning, LLM alignment, agentic systems, preference optimization, MPC, simulation-based evaluation.

Programming: Python, PyTorch, HuggingFace Transformers, scikit-learn, R, MATLAB, C/C++, SQL, Bash.

Systems: Git/GitHub, Linux, Docker, LaTeX, OpenAI-compatible APIs, EnergyPlus, experiment dashboards.

Languages: Mandarin Chinese (native), English (fluent), Cantonese (beginner).